JAR Z

HYDRAULIC OIL JAR, STRAIGHT PULL

JARRING

GENERAL DESCRIPTION

STRAIGHT-PULL HYDRAULIC OIL JAR FOR FISHING, DRILL STEM TESTING AND CORING. A METERING PISTON DELAYS THE STROKE WHILE THE OPERATOR TAKES STRETCH IN THE STRING, SO THE BLOW CAN BE SET FROM VERY LIGHT TO VERY HIGH IMPACT WITH NO ADJUSTMENT BEFORE OR AFTER THE RUN. THE HYDRAULIC SYSTEM IS CLOSED: OIL CANNOT ESCAPE AND WELL FLUIDS CANNOT ENTER. THE JAR RESETS WITH ONLY A FEW HUNDRED POUNDS OF WEIGHT AND TRANSMITS FULL TORQUE IN BOTH DIRECTIONS AT ALL TIMES.

TECHNICAL CHARACTERISTICS

Connection	O.D.	I.D.	Total stroke	Test pull load (lb)	Max. jarring load (lb)	Tensile at yield (lb)	Torque at yield (ft-lb)	Collar weight above jar (lb)
Sucker rod	1-5/8	1/4	7-1/16	5,000 - 8,500	15,400	46,300	260	1,100 - 1,450
1-13/16 WFJ	1-13/16	5/16	7	6,000 - 10,000	18,000	59,400	340	1,360 - 1,800
1-1/4 Reg	2-1/4	3/8	6-15/16	7,200 - 12,000	21,000	118,500	1,800	1,560 - 2,100
2-3/8 PH-6	2-29/32	1	11-7/16	12,600 - 19,000	35,400	194,800	2,260	2,200 - 3,000
2-3/8 EUE	3-1/16	1-1/2	11-9/16	12,000 - 16,000	27,800	160,200	2,200	2,300 - 3,100
2-3/8 Reg	3-1/8	1	11-13/16	15,000 - 23,000	32,400	229,200	4,060	2,400 - 3,300
2-7/8 Reg	3-3/4	1-1/4	11-13/16	18,000 - 26,000	56,500	345,000	7,640	4,200 - 5,700
2-3/8 IF	3-3/4	1-1/2	12-1/8	15,000 - 26,000	46,000	299,700	5,340	3,400 - 4,600
2-3/8 EUE	3-3/4	1-7/8	10-1/8	15,000 - 26,000	46,500	179,500	2,980	3,500 - 4,700
2-7/8 IF	4-1/4	1-15/16	11-13/16	18,000 - 28,000	46,700	430,300	9,920	3,500 - 4,700
2-7/8 EUE	4-1/2	2-3/8	12-1/4	18,000 - 28,000	49,000	375,000	11,160	3,600 - 4,900
3-1/2 FH	4-3/4	1-1/2	11-13/16	28,000 - 45,000	85,000	591,900	18,420	6,300 - 8,500
3-1/2 FH	4-3/4	2	13-5/16	28,000 - 45,000	74,500	468,800	17,200	5,600 - 7,500
4-1/2 FH	6	2	11-7/16	41,000 - 62,000	136,400	937,000	34,320	10,200 - 13,800
4-1/2 IF	6-1/4	2-1/4	14-3/8	55,000 - 80,000	159,000	917,400	40,680	11,800 - 16,000
5-1/2 Reg	6-3/4	2-3/8	14-11/16	62,000 - 110,000	172,800	1,013,800	48,660	13,000 - 17,500
6-5/8 Reg	7-3/4	3-1/16	14-3/8	63,000 - 100,000	149,000	1,587,900	64,020	11,000 - 15,000
7-5/8 Reg	9	3-3/4	13-13/16	80,000 - 120,000	214,000	1,621,000	92,260	14,300 - 19,600

Dimensions in inches. Tensile strengths are calculated theoretical yield values (about +/-20%); torque at yield is the theoretical value, recommended operating torque is 50% of it. Figures are a guide only: apply a safety factor.

DESIGN FEATURES

- **Adjustable blow:** impact set by the stretch taken in the string, no prior adjustment.
- **Closed hydraulic system:** internal parts are constantly lubricated; no leakage in or out.
- **Easy reset:** large piston ports open on closing, so only friction weight is needed.
- **Full torque:** transmitted in either direction at any point of the stroke.
- **Fast repeat blows:** limited only by the time to raise and lower the string.

OPERATING NOTES

- **Fishing:** run directly below the drill collars; add a jar intensifier about four collars above the jar.
- **Drill stem testing, coring:** 3 to 15 collars above the jar; does not interfere with test tools.
- **Never exceed the maximum jarring load.** Do not tong the threaded joints; tong at least 4 in away.
- **First blow:** can take seconds to minutes, depending on depth, stretch, temperature and hole condition.

HOW TO ORDER

Please specify: 1. jar size (O.D. x I.D.) 2. top and bottom connections, if not standard 3. spares: packing set, piston assembly, non-extrusion and seal protector rings, fill plugs 4. service kit and jar oil.